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Anomalous Spin Excitation in the Metallic State of $La_{2-x}Sr_xCuO_4$ Observed by Cu NQR

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We have measured the temperature dependence of the planar ^{63}Cu nuclear spin-lattice relaxation rate, $1/T_1$, up to 700 K in underdoped $La_{2-x}Sr_xCuO_4$ ($x = 0.08, 0.10, 0.12$). The observed $1/T_1T$ for those samples show Curie-Weiss like behaviour at high temperatures as previously reported. However, we can extract a characteristic temperature, T^*, below which the data deviate from the high temperature Curie-Weiss law for each sample. The hole concentration dependence of T^* is discussed from a viewpoint of the pseudo spin-gap in this single layer system. We have also found an anomalous behaviour in which the copper relaxation rate, $1/T_1$, depends on the NQR frequency especially in under-doping region. In this system, it is important to recognize microscopic disorder of the CuO_2 planes which modifies the intrinsic spin dynamics.

KEYWORDS: high-T_c superconductivity, $La_{2-x}Sr_xCuO_4$, NQR, spin-charge differentiation, pseudo spin-gap

§1. Introduction

The mechanism of high-T_c superconductivity is still an intriguing problem and many experiments have been carried out from various aspects. One of the best known characteristics of the high-T_c superconductors is the presence of CuO_2 planes in the crystal structures. Some materials contain a single plane per unit cell, *e.g.* $La_{2-x}Sr_xCuO_4$ (LSCO), $HgBa_2Ca_{n-1}Cu_nO_{2n+2+\delta}$, $Bi_2Sr_2Ca_{n-1}Cu_nO_{2n+4+\delta}$ and $Tl_2Ba_2Ca_{n-1}Cu_nO_{2n+4+\delta}$ with $n = 1$. The Hg, Bi and Tl cuprates with $n = 2$ or 3 have double or triple sets of planes in the unit cell, while $YBa_2Cu_3O_{7-\delta}$ (YBCO) has double planes. The two dimensional structure is believed to be closely related with the high-T_c superconductivity. Another prominent feature is that the normal metallic state of high-T_c superconductors have anomalous properties observed by various experiments.[1] Such anomalous properties are especially prominent in the underdoped region, which can not be understood by a simple Fermi liquid theory. These anomalous metallic states stem from the characteristic spin dynamics of low-dimensional systems near the Mott transition, and they have attracted much experimental and theoretical interest. With respect to spin dynamics, the spin excitation in multi-layered high-T_c superconductors near the Mott transition has been found to be strongly suppressed well above the superconducting transition temperature T_c.[2] We can define T_s as the temperature where $1/T_1T$ (T_1 is the Cu nuclear spin-lattice relaxation time) has a maximum value above T_c, and the electronic state from T_c to T_s is called the pseudo spin-gap state.[3]

To clarify the properties characteristic of the single layer CuO_2 plane, we have made an extended study of the nuclear spin-lattice relaxation rate, $1/T_1$, in LSCO using the planar Cu nuclear quadrupole resonance (NQR) technique. LSCO is a typical high-T_c superconductor and has the single layered structure in a unit cell. Furthermore, the adjacent CuO_2 planes perpendicular to the c-axis are separated from each other by La (Sr) O layers so that the interlayer coupling between the CuO_2 planes is thought to be small. This system is a hole-doped superconductor by cation substitution of La^{3+} by Sr^{2+}. The optimally doped sample ($x = 0.15$) has the T_c of about 38 K, and this T_c is rather low compared with that of the YBCO system ($T_c = 90$ K).

NMR and NQR techniques allow us to study magnetic excitations by examining the low frequency spin dynamics precisely. Previous results for LSCO below room temperature do not show clear indication of the pseudo spin-gap well above T_c, but the observed $1/T_1T$ follows the Curie-Weiss (CW) law below room temperature, similar to the other high-T_c cuprates.[4,5] From these results, it has been suggested that the pseudo spin-gap phenomenon stems from the bilayer coupling.[6,7] However, the origin of the pseudo spin-gap is still a controversial problem, because $1/T_1T$ of a single layered $HgBa_2CuO_{4+\delta}$ is highly suppressed below about 260 K.[8] Theoretically, there have been many attempts to explain these phenomena from models and dimensionality arguments.[9] From the perspective of the two dimensional properties of quantum Heisenberg antiferromagnets, the spin dynamics in high temperature region ($T > 0.5J$) is different from that of lower temperatures.[10-12] It was also pointed out that in LSCO, the relaxation rates of Cu do not depend on temperature, regardless of the samples are metallic or insulating.[12] However, a subsequent study revealed the appreciable differences of $1/T_1$ at high temperatures between La_2CuO_4 and $La_{2-x}Sr_xCuO_4$ ($x = 0.15$).[13]

We have made measurements of $1/T_1$ using a conventional NQR apparatus in underdoped LSCO samples ($x = 0.08$, 0.10, and 0.12), and compared the results with the previously measured $1/T_1$ of lightly doped samples ($x = 0.06$, 0.15, 0.20).[14-16] We found the anomalous behaviour of $1/T_1$ depending both on the NQR frequency

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and the temperature. Our purpose in this paper is to give an overview of the spin dynamics in the normal state of the LSCO, and it is composed as follows. In §2, we show the experimental procedure and results. The Cu-NQR frequency spectra and newly found frequency dependence of Cu-nuclear spin-lattice relaxation rate are shown here. The properties of $1/T_1$ in the high temperature region and indication for the existence of pseudo spin-gap phenomena are also discussed. Section 3 is assigned to the concluding remarks.

§2. Experimental Results and Discussion

2.1 Sample preparation and experimental technique

Powdered samples of LSCO with $x = 0.06 \sim 0.20$ were prepared by standard solid-state reactions of CuO, La_2O_3 and $SrCO_3$. All the samples were confirmed by X-ray diffraction (XRD) to be a single phase. We determined T_c from the change in coil inductance of an NQR probe for each sample, and the results were 7 K ($x = 0.06$), 22 K ($x = 0.08$), 30 K ($x = 0.10$), 31K ($x = 0.12$), 38 K ($x = 0.15$) and 30 K ($x = 0.20$). The NQR experiments were performed by a conventional pulsed NQR spectrometer with a variable temperature control system.

2.2 Frequency spectra

The zero field Cu-NQR spectra of LSCO ($x = 0.06 \sim 0.15$) are shown in Fig. 1. All the spectra can be decomposed into two Cu NQR lines associated with charge differentiated Cu sites, A and B, and each site has two peaks corresponding to ^{65}Cu- and ^{63}Cu–NQR lines.[17,18] Although the origin of these two distinct sites has been discussed from various standpoints, this feature is not understood clearly.[17-19]

The Cu-NQR frequency, ν_Q, is determined by the nuclear quadrupole moment Q and the electric field gradient (EFG) at the Cu site, q, as;[20]

$$\nu_Q = \frac{e^2qQ}{2h}\left(1+\frac{1}{3}\eta^2\right)^{1/2},$$

where η is the asymmetric parameter of the EFG. We can determine ν_Q for each LSCO sample from the observed peak frequency of Cu-NQR. The peak frequency of the A-line of ^{63}Cu increases linearly in proportion to the strontium concentration.[17] This is similar to the YBCO system in which the peak frequency increases in proportion to the oxygen content.[18] In contrast, the dependence from the NQR line width on strontium concentration is very weak over the range from $x = 0.06$ to $x = 0.15$. Previously reported ^{63}Cu-NQR spectra of the parent material, La_2CuO_4 have an FWHM (full width at half maximum) of about 60 kHz.[12] If the NQR line widths were affected mainly by the lattice distortion induced by substitution, the FWHM might increase in proportion to the strontium concentration. The observation contrasts with this assumption. Therefore, we have to consider other factors related with widely distributed EFG.

In this system, only a slight substitution of strontium ($x \sim 0.06$) for lanthanum makes a superconducting

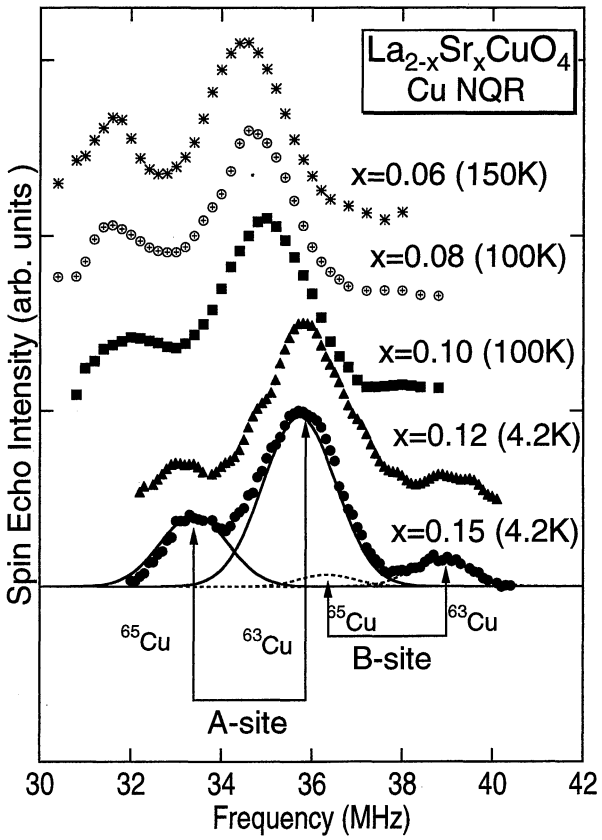


Fig. 1. Cu NQR frequency spectra of LSCO ($x = 0.06 \sim 0.15$). All spectra can be decomposed into two distinct charge differentiated Cu sites shown as A and B.

phase, and at the same time, the FWHM of the NQR spectra becomes about 1.9 MHz. This is more than 30 times broader than that of La_2CuO_4. Thus, the drastic change of the NQR line width occurs between the $x = 0.00$ and $x = 0.06$ samples which corresponds the metal-insulator transition. Therefore, we can associate the wide distribution of EFG q with the doped carrier. This also suggests the doped conduction carriers have a key role to interpret the observed spectra.

In general, EFG q can be divided into two parts as $q = q_{\rm latt}(1-\gamma_\infty) + q_{\rm loc}(1-R_Q)$,[21,22] where $q_{\rm latt}$ and $q_{\rm loc}$ are the electric field gradients arising from the lattice ions external to the Cu site and the contributions from conduction electrons within the atomic sphere, respectively. The γ_∞ and R_Q represent Sternheimer antishielding factor and shielding factor. In high-T_c cuprates, it is suggested that the anisotropic hybridization between Cu-3d, 3p, 4p and neighbouring O-2p electrons is strong. This anisotropy of the charge density around the Cu atom sensitively affects the EFG inside each Cu atom ($q_{\rm loc}$).[22] The experimental results show the broad NQR spectra which are characteristic of the metallic nature of the LSCO system ($x > 0.05$) as mentioned above. This suggests that the charge density in each Cu orbital is inhomogeneously distributed from the NQR aspect.

2.3 Frequency dependence of $1/T_1$

The observed broad NQR line width of ^{63}Cu(A) NQR, about 1.9 MHz FWHM even for the $x = 0.06$ low-doping

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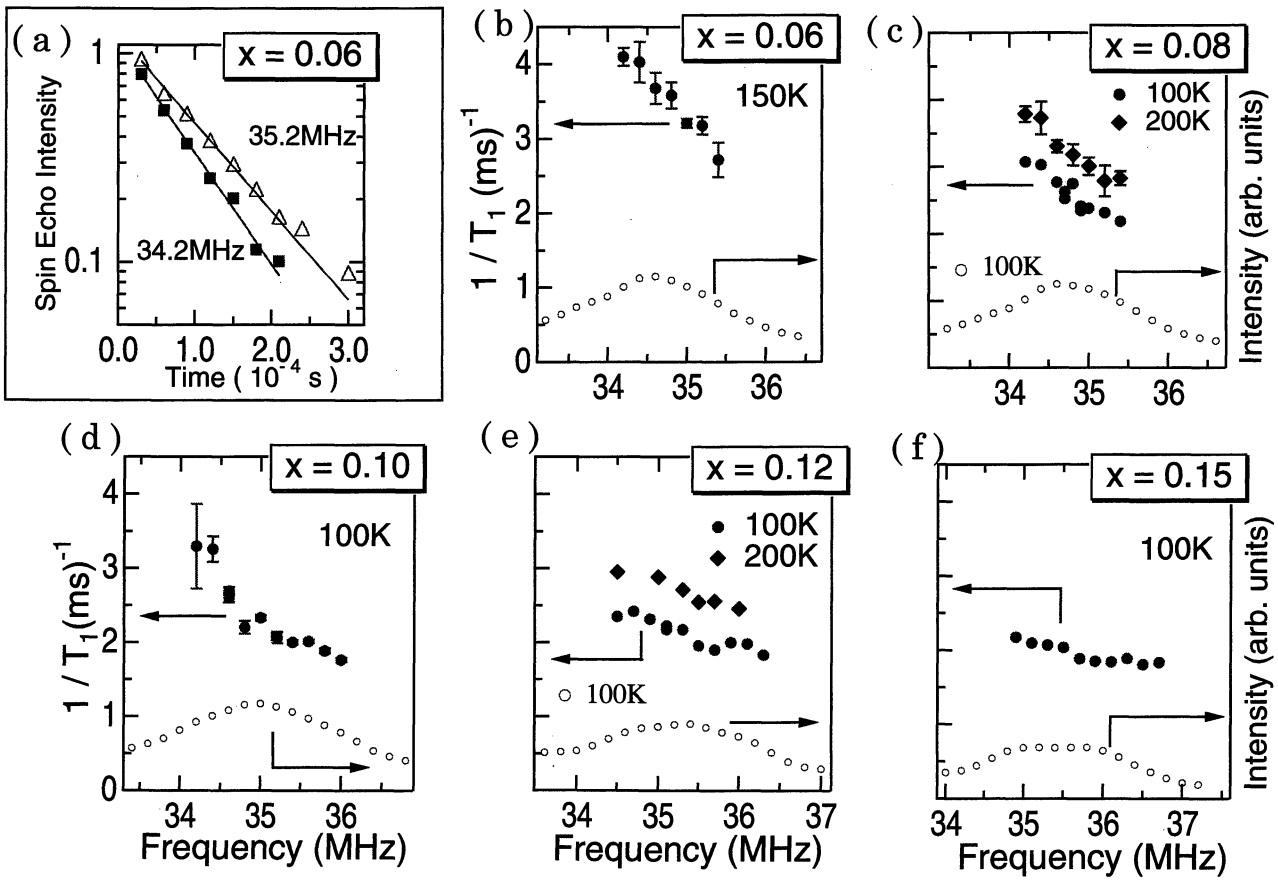


Fig. 2. (a) The typical recovery curves, $[M(\infty) - M(t)]/M(\infty)$, for $x = 0.06$ at centered frequency 34.2 and 35.2 MHz at $T = 150$ K. (b)–(f) Frequency dependence of ^{63}Cu relaxation rate $1/T_1$ around the peak frequency of the A-site. The opened circles show relative intensity of Cu-NQR for $x = 0.06 \sim 0.15$.

sample, is rather unusual because the homogeneity of sample has been ensured by XRD. It raises two following questions. Does the distributed EFG indicate the locally distributed magnetic states? Or is the spin excitation state uniquely identified for each sample? To explore these points, we have intensively measured the frequency dependence of $1/T_1$ across the inhomogeneously distributed spectra.

The results are shown in Fig. 2. In the measurement of the NQR frequency dependence of $1/T_1$, the time width of the saturating pulse was kept at a constant value of 1.8 μs. This pulse width corresponds to 140 kHz in the frequency domain. This is a narrow frequency width compared with that of NQR spectra (about 1.9 MHz). Although the interference of the B-site of ^{65}Cu can not be completely ruled out, this effect can be considered to be relatively small in lower doping region, as is expected from the NQR spectra shown in Fig. 1.

The results shown in Fig. 2 exhibit the following two remarkable features. (1) As the frequency increases from that of the peak position of ^{63}Cu, the relaxation rate, $1/T_1$ becomes progressively smaller. (2) This tendency is more pronounced in underdoped region. In the optimal doped sample ($x = 0.15$), this frequency dependence of $1/T_1$ is hardly seen within the experimental accuracy. If the frequency dependence of $1/T_1$ observed here were caused by the interference of the B-site, it should be more pronounced for larger strontium concentration. The observation of (2) rules out this possibility. Thus, this frequency dependence is an intrinsic anomaly in the underdoped region of the LSCO system. This may be related to inhomogeneity of the concentration of doped holes in the CuO_2 planes. However, at present, we have no idea to explain the facts (1) and (2) in a convincing way.

2.4 Nuclear spin-lattice relaxation rate in the high temperature region

We have measured the ^{63}Cu nuclear spin-lattice relaxation rate $1/T_1$ of underdoped LSCO ($x = 0.08$, 0.10 and 0.12) up to 700 K. In this experiment, we directed our attention to the temperature dependence of ν_Q. We adjusted the rf-frequency of the apparatus to the peak frequency of the ^{63}Cu of the A-site at each temperature so as to avoid the effect of the frequency dependence of $1/T_1$ discussed in §2.3. The results are shown in Fig. 3, and compared with other available data for $x = 0.06$,[14] 0.15,[15] and 0.20.[16] We made use of an oven in which the temperature kept almost constant within 2 K. After this experiment, we could not have any indication that heated samples changed in quality.

In the high temperature region, the observed $1/T_1$ depends on temperature for each sample, and the properties of temperature dependence are different from each other. The $1/T_1$ make maxima at about 200–300 K for

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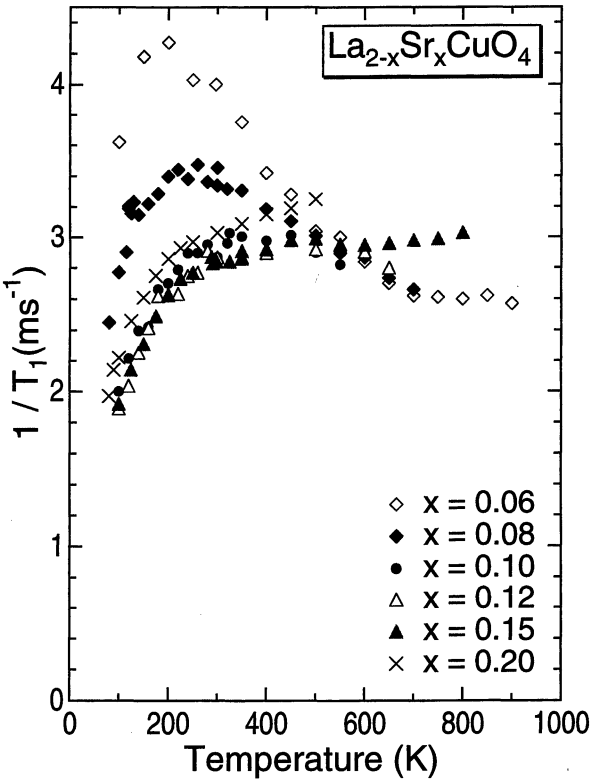


Fig. 3. ^{63}Cu relaxation rate $1/T_1$ (defined by $2W$) of underdoped LSCO. The data of $x = 0.06$, 0.15 and 0.20 are reproduced from refs. 14–16.

$x = 0.06$, 0.08, 0.10 samples, and decrease in proportion to the temperature. On the other hand, the $1/T_1$ for more strontium substituted samples make no maxima. These features make contrasts with the results as were argued by Imai *et al.*[12] If exchange-coupled local moments in the high temperature region ($T > 0.5J$) grow, the expected $1/T_1$ is described within the Gaussian approximation as,[23]

$$\frac{1}{{}^{63}T_{1\infty}} = \sqrt{2\pi}(A_x^2 + 4B^2)\frac{S(S+1)}{3\hbar^2\omega_e'}$$

where

$$(\omega_e')^2 = \frac{2zJ^2}{3\hbar^2}S(S+1)\left[1 - \frac{2A_xB}{A_x^2 + 4B^2}\right].$$

We make use of the derived values of J, A_x and B from the Cu nuclear spin-lattice relaxation rate experiment of La_2CuO_4.[12,13] If we take $S = 1/2$, $z = 4$, $J = 1600\,\mathrm{K}$, $A_x = 40\,\mathrm{kOe}/\mu_\mathrm{B}$ and $B = 40\,\mathrm{kOe}/\mu_\mathrm{B}$, a temperature independent value of $1/T_1$ of 4200 (s^{-1}) is obtained. Our experimental value of $1/T_1$ ($x = 0.08$) at 700 K is about 2600 (s^{-1}), and thus the calculated relaxation rate is higher than the experimental values by about a factor of 1.6. In addition, the observed $1/T_1$ depends on temperature, and the temperature dependence of each sample is apparently different each other. Therefore, the local moments at high temperatures are not likely in this system.

Curie-Weiss behaviour of $1/T_1T = C/(T + \theta)$ has been experimentally observed at high temperatures ($T > 300\,\mathrm{K}$).[15] Two parameters C (the Curie constant) and θ

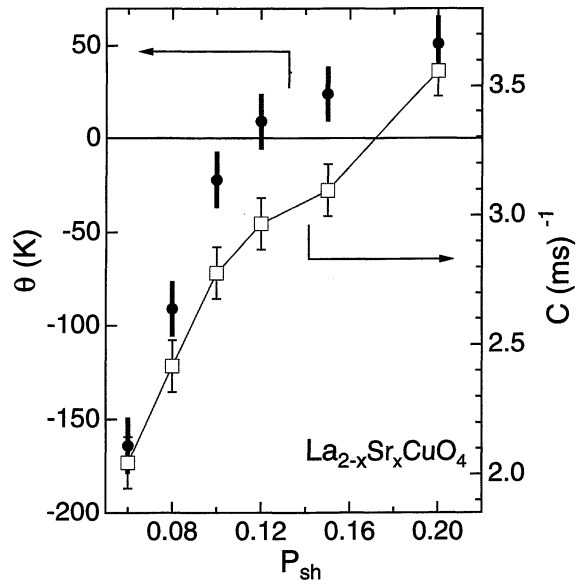


Fig. 4. Parameters fitted to the observed $1/T_1T$ above T^* by CW law plotted against the number of doped holes per a CuO_2 sheet (●: Weiss temperature (K), □: Curie constant (s^{-1})).

(the Weiss temperature) were determined for each LSCO sample as shown in Fig. 4. This analysis yields a direct information of the spin dynamics in the high temperature region, in which can be compared with theoretical calculations based on two dimensional antiferromagnetic spin fluctuations. Two different theoretical approaches to this problem are the self-consistent renormalization (SCR) treatment for Fermi liquid and the mean field approximation based on the t-J model.[24-28] We made here a qualitative comparison with the experimental results.

The Weiss temperatures θ derived from a Curie-Weiss fit increase continuously from $-150\,\mathrm{K}$ (for $x = 0.06$) to 50 K (for $x = 0.20$) with the hole concentration. Moreover, θ approaches 0 K around the optimal doping region ($x \approx 0.15$). This inclination is reproduced qualitatively by the two theoretical calculations.[24-26] However, the observed doping dependence of the Curie constant C is clearly different from the mean field calculations.[26-28] Within the assumption that J is not dependent on the hole doping, the caluculated Curie constants are almost independent of it.[28] Whereas the experimentally derived values of C increase in proportion to the strontium concentration.

2.5 Suppression of antiferromagnetic correlation

All the $1/T_1T$ data can be fitted to the CW law (namely, T_1T shows a linear relation with the temperature) above room temperatures, but start to deviate suppressively below a characteristic temperature T^* from the fitted curve as is shown in Fig. 5.

Here, we discuss the hole doping dependence of T^*. The extracted T^* is shown as a function of the number of doped holes in a CuO_2 plane ($P_\mathrm{sh} = x$ in the ionic model)[29] in Fig. 6, together with the previously reported data.[14-16] The T^* decreases monotonically as the number of doped holes to the CuO_2 planes increases toward the optimal doping.[3,30] This trend is similar to

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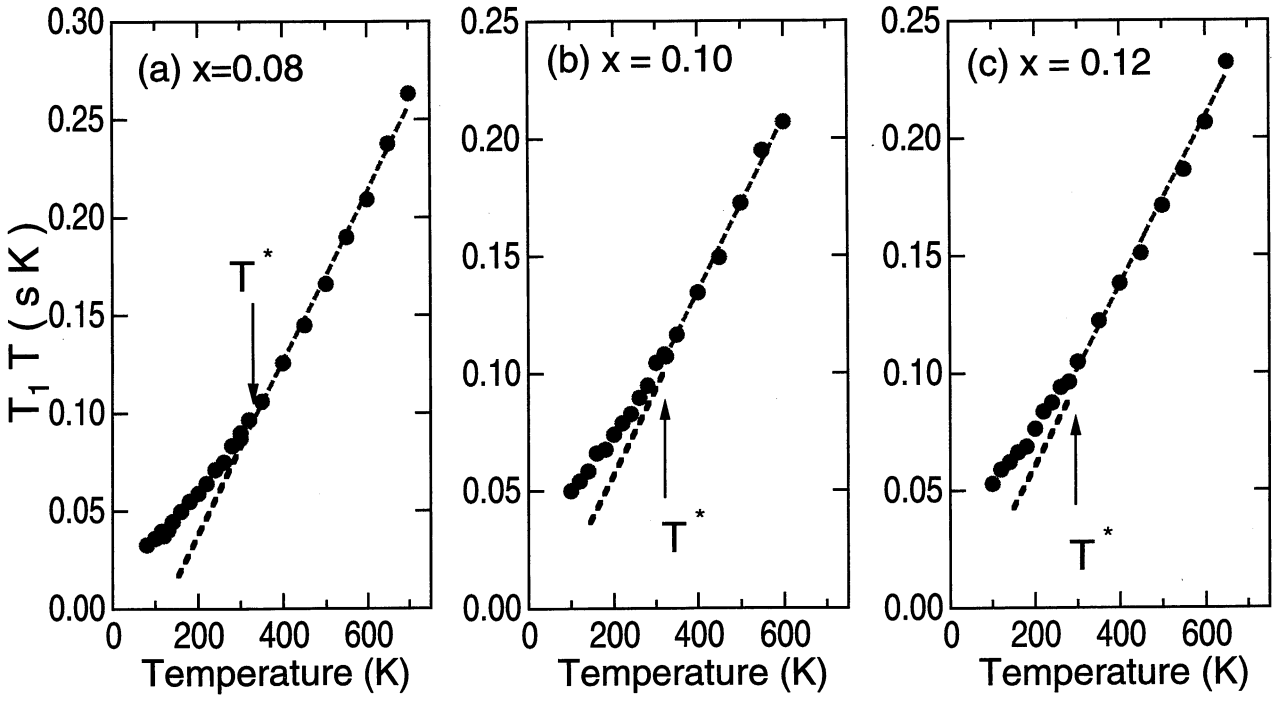


Fig. 5. (a)–(c) The observed T_1T of $x = 0.08$, 0.10 and 0.12. The dashed line is obtained from a least squares fit. The arrow indicates the onset temperature for deviation from the fitted CW law, T^*.

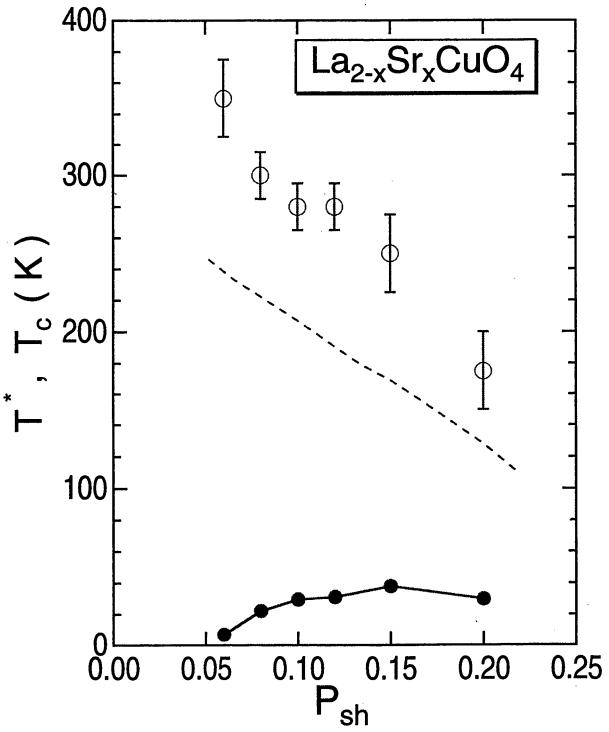


Fig. 6. Experimentally determined T^* (open symbol) and T_c (closed symbol) of LSCO ($x = 0.06 \sim 0.20$) plotted against the number of doped holes per a CuO_2 sheet. The dashed line shows a reference line from numerically calculated temperatures of singlet RVB formation when $J = 1600$ K (ref. 28).

the properties of the pseudo spin-gap observed in the underdoped YBCO system.[31] To clarify the doping dependence of T^*, we compare the observed values with theoretical results. Theoretically calculated T_1T increases in proportion to the temperature, and looks like the CW like behaviour in the high temperature region.[24-26] Besides, the T_1T tends to saturate at lower temperatures, and then we can define theoretical T^* for each hole doping. However, the calculated doping dependence of T^* shows the opposite feature with that of the experimental T^*.[8]

Here, we notice the mean-field calculations of the extended t-J model based on the singlet RVB (resonating valence bond) state.[27,28] This calculation suggests the opening of a $d_{x^2-y^2}$ wave pseudo gap in spin excitation spectrum below the singlet RVB formation temperature, T_{RVB} ($T_{RVB} > T_c$), in underdoped region. Theoretical T_{RVB} decreases as the number of doped holes increases. This is the same inclination as that of the experimental T^*. Moreover, the experimental $1/T_1T$ in the overdoped $x = 0.20$ sample also show CW like behaviour with only a little deviation at 180 K.[16] The qualitative agreement of the doping dependence between the calculated T_{RVB} and the experimental T^* suggests that the suppression of antiferromagnetic correlation observed here may be associated with an onset temperature of the pseudo spin-gap opening.

A sharp peaking in $1/T_1T$ is observed around 50 K just above T_c for the ^{63}Cu-enriched LSCO ($x = 0.13$ and 0.18).[5] From the anisotropy of T_1, the peaking of $1/T_1T$ just above T_c has been interpreted either as an unusual rapid reduction of the antiferromagnetic correlation length or as the pseudo spin-gap opening at higher temperature.[5] One should note that such an anomalous behavior of T_1 just above T_c appears in the low-temperature state below T^* with the unconventional sup-

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pression of the antiferromagnetic correlation. Here, we also found the frequency dependent T_1 indicative of inhomogenous electronic states. Thus, one has to take into consideration some kind of inhomogeneity to understand the unconventional spin dynamics below T^*. However, at the present stage, it is not easy task.

§3. Conclusion

In typical high-T_c superconducting cuprates, $La_{2-x}Sr_xCuO_4$, which have a single CuO_2 layer in a unit cell, the anomalous suppression of $1/T_1T$ was observed by extended high temperature measurements. These phenomena are associated with the pseudo spin-gap phenomena as other bilayer superconductors, and describe anomalous spin dynamics in the CuO_2 plane of underdoped high-T_c superconductors. However, this system exhibits an extremely inhomogeneous metallic state caused by the strontium substitution with lanthanum. This feature, characteristic in this system, affects the spin excitation qualitatively and may be related with the problem why the $1/T_1T$ does not show any broad maximum above T_c as observed in a single layered $HgBa_2CuO_{4+\delta}$ system.[8] We also found the NQR frequency dependence of $1/T_1$, which is remarkable especially in low doping region. Our experiments suggest the underdoped LSCO has some kind of inhomogeneous electronic state. This spatial inhomogeneity is not negligible to understand the intrinsic spin dynamics of the LSCO system.

Acknowledgments

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